
BIOLOGY

9700/33

Paper 3 (Advanced Practical Skills 1)

May/June 2016

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9700	33

Mark scheme abbreviations:

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
<u>underline</u>	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9700	33

- 1 (a) (i) (*risk assessment*)
(hydrogen peroxide) harmful **or** irritant + medium **or** high ; [1]
- (b) (i) (*measures room temperature*)
whole number **or** to half a degree + °C ; [1]
- (ii) (*decides on interval for temperature*)
at least three additional temperatures + whole numbers + even intervals ;
°C ; [2]
- (iii) (*recording results*)
1. table drawn + heading, temperature + °C ;
2. heading, time + seconds ;
3. records results for at least five temperatures ;
4. correct pattern of results ;
5. times recorded as whole seconds ;
6. records results for repeats + means calculated ; [6]
- (iv) (*source of error with reason*)
appropriate error with reason ;
e.g. concentration of hydrogen peroxide decreases
appropriate error with reason ;
e.g. different volumes of extract on each square of filter paper [2]
- (v) (*conclusions*)
(as temperature increases, activity increases) more successful collisions **or**
more enzyme-substrate-complexes/ ESCs ;
(decreased / no activity) denatures **or** changed shape of active site ; [2]
- (vi) (*modification to investigate another variable*)
1. (to standardise temperature) stated temperature + thermostatically
controlled water-bath ;
2. (independent variable) at least five concentrations of catalase ;
3. (method) simple dilution / proportional dilution / serial dilution ; [3]
- (c) (*chart*)
1. (x-axis) different plant species + (y-axis) initial rate of activity of catalase /
s⁻¹ ;
2. (scale on x-axis) even width of bars + (scale on y-axis) 0.05 to 2 cm, labelled
at least each 2 cm ;
3. correct plotting of five bars ;
4. five bars labelled with each horizontal line drawn as a thin line + each
column labelled ; [4]

[Total: 21]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9700	33

- 2 (a) (i) (*plan diagram*)
1. plan diagram of appropriate size + no shading ;
 2. no cells + at least two vascular bundles + correct section drawn ;
 3. epidermis drawn as two lines drawn closely together ;
 4. line drawn to show area of cells located at tip of leaf ;
- [4]
- (ii) (*drawing*)
1. quality of line for outer wall of cells + size at least 50 mm across largest cell ;
 2. only four cells drawn, each cell touching at least one other cell ;
 3. cell walls drawn as two lines close together ;
 4. one cell which shows a difference from other cells ;
e.g. cell contains an inclusion
 5. uses one label line + one label to cell wall ;
- [5]
- (b) (i) (*ratio*)
1. measures depth of midrib + diameter of the vascular bundle ;
 2. records whole numbers **or** to 0.5 for both measurements ;
 3. *decides to* use same units for both measurements ;
 4. displays, in final ratio, larger number to smaller number ;
 5. final answer as simplest ratio ;
- [5]
- (ii) (*conclusion*)
- (habitat) water + (feature) large air spaces **or** more air spaces **or** AVP ;
- [1]
- (c) (*observable difference between leaf on K1 and leaf in Fig. 2.2*)
- organises comparisons into three columns with one column for features, one headed **K1** and one headed **Fig. 2.2** ;
- any *three* observable differences of comparison ;;;
- e.g. **K1** has more vascular bundles than **Fig. 2.2**
- [4]

[Total: 19]